Band-1 Hadamard Order 2836 — Miyamoto (H_708 Paley-II); Miyamoto-Only Bonus; Gap 2052 Still OPEN

H_708/C_710/H_2836 PASS (Miyamoto); 2836 NOT in SPK (Paley-I/II/Kronecker FAIL); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_2804_Miyamoto`; `2026-09-15_Band1_Hadamard_2708_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_2836_miyamoto/`.
Cutoff. 2026-09-15 (~02:48Z).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 708 and 2836 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-II seed H_{708}. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 2836 is a **Miyamoto-only bonus** (**not** in SPK): honest SPK audit — Paley-I **FAIL** ($q = 2835$ not prime); Paley-II **FAIL** ($q = 1417$ not prime); Kronecker **FAIL** ($2836 = 4 \cdot 709$, 709 not Hadamard-admissible). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 2804 on box. This increment constructs the **next preferred** Miyamoto-eligible target after $q = 701$: **order 2836** ($2836 = 4 \cdot 709$, 709 prime $\equiv 1 \pmod 4$). Seed H_{708} via Paley type II ($q = 353$ prime $\equiv 1 \pmod 4$; order $2(q+1) = 708$). Symmetric Paley conference C_{710} over prime 709. $H_{2836} = \text{Miyamoto}(H_{708}, C_{710})$. Unit tests **26/26 PASS**. Order 2836 does **not** enter SPK (Miyamoto-only bonus). SPK next gap remains **2052** (OPEN). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 2836; Miyamoto; Paley type II; Paley conference; Miyamoto-only bonus; constructive gap 2052; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{708} constructed and verified (Paley type II, $q = 353$). 2. C_{710} constructed and verified (Paley conference, $q = 709$): $CC^\top = 709I$, zero diagonal. 3. H_{2836} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 4. SPK: $2836 \notin$ SPK (Miyamoto-only bonus). Paley-I, Paley-II, and Kronecker paths for 2836 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$2836 = 4 \cdot 709$
$q = 709$	prime, $709 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{708}
Needed conference	order $710 = q + 1$
H_{708} path (used)	Paley type II, $q = 353$ (prime $\equiv 1 \pmod 4$); $2(q+1) = 708$
Conference path	Symmetric Paley over prime 709
Paley-I for 2836	**REJECTED**: $q' = 2835$ not prime
Paley-II for 2836	**REJECTED**: $q' = 1417$ not prime
Kronecker for 2836	**REJECTED**: only 4×709; $709 \not\equiv 0 \pmod 4$ (not admissible)
SPK membership	**False** (Miyamoto-only bonus)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{708}	Paley type II, $q = 353$	0	0.2197	**PASS**
C_{710}	Paley conference, $q = 709$	0	0.3990	**PASS**
H_{2836}	Miyamoto(H_{708}, C_{710})	0	9.8933	**PASS**

Primary Miyamoto pipeline wall (incl. unit tests + SPK refresh) $\approx$ 10.9487 s. Compressed ± 1 matrices saved as `H708_pm1.npz`, `C710_pm1.npz`, `H2836_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior ships (incl. 700, 2804) $\cup$ verified $H_{708} \cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 2836 therefore remains **outside** SPK.

Quantity	Value
Known count	882
2836 in SPK	**False** (Miyamoto-only bonus)
708 in SPK	True (Paley II $q = 353$)
2804 in SPK	True (prior Paley I)
2692 in SPK	False (prior Miyamoto bonus)
Next standard gap	**2052** (OPEN)
Missing first (sample)	2052, 2060, 2076, 2092, 2108, 2148, 2156, 2172, 2212, 2284…

5. Gap 2052 (still OPEN; non-finding ONLY)

This ship does **not** fill gap 2052. Prior triage stands: Paley-I/II / Kronecker / Miyamoto-Paley-conf all **FAIL** for order 2052. Claim level for 2052 remains method triage / non-finding ONLY. Non-finding $\neq$ counterexample; Hadamard conjecture remains OPEN. No H_{2052} matrix is claimed.

6. Unit tests (26/26)

Test	Result
`H4_sylvester`	PASS
`H20_paley_I_control`	PASS
`paley_II_q5_H12`	PASS
`paley_II_q13_H28`	PASS
`miyamoto_H20_from_H4_conf6`	PASS

Test	Result
paley_I_q11_H12	PASS
q353_is_prime	PASS
q353_equiv_1_mod_4	PASS
H708_paley_II_order_formula	PASS
q709_is_prime	PASS
q709_equiv_1_mod_4	PASS
n2836_equals_4_times_709	PASS
H708_order_formula	PASS
C710_order_formula	PASS
paley_I_H2836_path_rejected	PASS
paley_II_H2836_path_rejected	PASS
kronecker_H2836_path_rejected	PASS
q2835_not_prime	PASS
q1417_not_prime	PASS
reject_all_ones_H8	PASS
conference_6_paley_control	PASS
miyamoto_eligible_2836_arith	PASS
conference_710_shape_expected	PASS
gap_2052_not_filled_by_this_ship	PASS
2836_miyamoto_only_bonus_arith	PASS
paley_II_seed_pattern_control	PASS

7. Next optional targets

Miyamoto-eligible primes $q > 709$, $q \equiv 1 \pmod 4$ (sample):

q	$n = 4q$	Need H_{q-1}	On-box path notes
733	2932	732	—
757	3028	756	—
761	3044	760	Double(PaleyI half=380), Kronecker(380x2)
769	3076	768	Double(PaleyI half=384), Kronecker(4x192)
773	3092	772	—
797	3188	796	PaleyII(q=397)
809	3236	808	Kronecker notes
821	3284	820	PaleyII(q=409)

Preferred next. Method-attack remaining SPK gap **2052** (GS/SDS for order ~513 if literature seeds appear), **or** construct Miyamoto $q = 761 \to n = 3044$ (seed Double/Kronecker notes), **or**

$q = 797 \to n = 3188$ (Paley-II $q = 397$), **or** Track B privacy / Object 1 meter await. GATE-04 remains LOCKED.

8. Non-claims (explicit)

- Not the full Hadamard conjecture.
- Not Clay-closed / finished-ToE / RH / P≠NP / YM / Frankl / NS-solved / AC-solved.
- Gap **2052** remains **OPEN**.
- Order 2836 is **not** in SPK (Miyamoto-only bonus).
- GATE-04 collection **LOCKED**; no invented J.
- `clay_closed = false`.

9. Artifacts

Lab: `/workspace/experiments/band1/hadamard_2836_miyamoto/`
Verifier: `verify_hadamard_2836.py`
Results: `results.json`, `literature_ledger.json`, `SUMMARY.txt`
Matrices: `H708_pm1.npz`, `C710_pm1.npz`, `H2836_pm1.npz`
Shelf: `/workspace/DOWNLOAD_PDFs/2026-09-15_Band1_Hadamard_2836_Miyamoto.pdf`